# A Governance-First Architecture for AI-Assisted Qualitative Research: Design Principles and Open Infrastructure for Responsible Human–AI Collaboration

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### Abstract

The proliferation of large language models as analytical tools in qualitative research has arrived faster than the governance frameworks needed to deploy them responsibly. Existing platforms prioritise capability while leaving the questions that define whether an analysis is credible — reflexivity, auditability, data protection, evidential grounding — for the researcher to solve manually. This paper describes a governance-first architecture for AI-assisted qualitative research: a modular, open-source pipeline in which methodological safeguards are enforced at the infrastructure level rather than delegated to researcher self-discipline. The contribution is architectural rather than algorithmic: the pipeline specifies a governance infrastructure within which AI-assisted analysis can be conducted responsibly, regardless of which underlying model is used. The architecture achieves this through a two-pass analysis structure with cryptographic anchoring to prevent post-hoc revision, a structured researcher reflexivity elicitation module locked before the second pass, DPIA gating as a hard pipeline constraint for special-category data, per-claim grounding verification with human sign-off blocking synthesis, and an automatically generated authorship disclosure record. Argued here is that enforcing these safeguards programmatically — rather than recommending them procedurally — constitutes a meaningful advance in what responsible AI-assisted qualitative research can look like. The pipeline is presented as an untested design contribution; the repository is publicly available for inspection and empirical evaluation.

**Keywords:** AI-assisted qualitative research, research governance, reflexivity, audit trail, GDPR, open science, large language models, human-AI collaboration, research infrastructure

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## Introduction

There is a particular kind of frustration familiar to anyone who has done extensive qualitative work: the long evening of transcript reading, the growing pile of margin

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notes, the slowly emerging sense that something important is in the data but the sheer volume of it is preventing you from seeing the full shape of what is there. It is not a failure of method or of intelligence — it is a capacity problem, and it has always been one. The emergence of large language models capable of reading, summarising, and annotating text at scale has made that problem look soluble in a way it has never been before. The question is whether it is actually soluble, or whether the speed and apparent fluency of AI-assisted analysis conceals new methodological problems even as it addresses old logistical ones.

This paper argues that both things are true simultaneously, and that the resolution lies in how the AI tools are deployed rather than in whether they are deployed at all. The current landscape of AI-assisted qualitative research tools is dominated by commercial platforms — CoLoop, the AI layers built into NVivo 15 and ATLAS.ti 25, MAXQDA AI Assist, and a growing number of standalone services — that make a broadly similar architectural choice: data is uploaded to cloud infrastructure, an LLM processes it, and coded or themed output is returned to the researcher. These tools are capable, and for many research contexts they represent a genuine improvement on purely manual approaches. But they share a structural limitation that becomes critical when the data is sensitive, the researcher is epistemologically committed to reflexive approaches, or the study is subject to formal ethical and legal scrutiny. The governance questions — who is responsible for what the AI produces, how the researcher's own positionality interacts with the analysis, what happens to participant data on external servers, and how a reader or examiner can verify that the analysis was conducted as described — are left entirely outside the tool.

This paper describes a different approach. The ‘Human Agentic-AI Adaptive Research Pipeline’ (HAAARP) presented here begins from the governance requirements and works backwards to the implementation, rather than the other way around. The result is a modular, open-source architecture in which the methodological safeguards that qualitative researchers already recognise as essential — reflexivity, auditability, pre-registration consistency, evidence grounding — are enforced at the infrastructure level. The pipeline cannot proceed to the second pass of analysis until the first has been cryptographically locked. It cannot ingest special-category data until a completed Data Protection Impact Assessment has been signed and recorded. It cannot synthesise findings until a human researcher has reviewed, adjudicated, and signed off on every claim the AI has made. These are not optional features or best-practice recommendations. They are the architecture. The contribution of this paper is architectural rather than algorithmic: it specifies a governance infrastructure within which AI-assisted analysis can be conducted responsibly, not a new analytical method or a superior model configuration. Like any AI-assisted method, the pipeline does not eliminate epistemic risk; it attempts to make those risks visible and auditable.

This is presented as a design contribution rather than an empirical one. The pipeline is publicly available as an open-source repository, documented in sufficient detail for inspection and replication, but it has not yet been evaluated in a live research study.

This is explicitly stated since as a design paper it is making different claims than an evaluation paper, and the distinction should be visible to the reader. The contribution is in the architecture itself — the argument that these governance constraints, enforced programmatically, constitute what would be a meaningful advance in how responsible AI-assisted qualitative research could look, regardless of which LLM is doing the actual reading.

## The Problem is Governance, Not Capability Alone

The debate about AI in qualitative research has sometimes proceeded as though the central question is whether LLMs can produce reliable qualitative codes — whether the output of a language model resembles what a trained researcher would produce. This is an interesting empirical question, and there is now a reasonable body of evidence suggesting that for certain tasks, particularly the initial categorisation and thematic grouping of open-ended survey or interview text, LLMs perform comparably to human coders (Tai et al., 2024; Ziems et al., 2024). But treating capability as the central question systematically displaces the questions that methodologists should be asking first.

The trustworthiness of qualitative research does not rest primarily on whether the coding was performed by a human or a machine. Lincoln and Guba's (1985) foundational framework of credibility, transferability, dependability, and confirmability asks, in effect, whether the analysis is transparent, consistent, and grounded in the data as it was encountered. These criteria translate to a set of operational requirements: there must be an audit trail showing what was analysed and when; the researcher's positionality must be acknowledged and its effect on the analysis must be traceable; analytical claims must be linked to specific evidence in the data; and the process must be documented in sufficient detail that a competent reader could evaluate it independently. None of these requirements are incompatible with AI assistance. But current tools do not support them — they produce output without provenance, and the researcher is left to reconstruct the methodological record manually after the fact.

The reflexivity problem is particularly acute. Reflexive approaches to qualitative analysis, including reflexive thematic analysis (Braun & Clarke, 2006; 2019), interpretive phenomenological analysis (Smith, Flowers, & Larkin, 2009), and psychodynamic discourse analysis, are not simply analytical methods — they are epistemological commitments to the view that the researcher's subjectivity is a constitutive element of the analysis rather than a source of bias to be corrected. The researcher's theoretical orientation, clinical experience, prior encounters with similar material, and working hypotheses all shape what gets noticed and how it gets named. A responsible reflexive analysis does not pretend otherwise; it documents that shaping process explicitly so that readers can evaluate what the interpretive lens contributed. Handing the analysis to an AI tool that has no knowledge of the researcher's

positionality does not make the analysis more objective — it makes the positionality invisible, which is precisely the opposite of what reflexive methodology requires.

The data governance problem is less philosophically complex but practically more serious for many researchers. A substantial proportion of qualitative research in health, psychology, social work, and education involves sensitive personal data that falls under GDPR Article 9 special category protections in the UK and EU: health information, mental health data, therapy session recordings, data about children. Submitting this material to a commercial cloud-based AI service is not simply a privacy concern in the colloquial sense — it may constitute a lawful processing violation if the relevant consents and data protection impact assessments do not cover that use. The researchers who most need the efficiency gains that AI assistance could provide — doctoral students conducting interviews with vulnerable populations, small teams in under-resourced clinical settings, solo practitioners with limited analytic support — are precisely those for whom the cloud-processing model is most difficult to use responsibly.

There is growing recognition in the literature that this gap exists. Van Dis et al. (2023) argue that the responsible use of generative AI in research requires institutional frameworks that go well beyond the terms of service of individual tools, and identify data privacy as among the most pressing unresolved concerns. Ashwin et al. (2022), evaluating AI-assisted analysis of interview data from displaced populations, found that LLM-generated annotations introduced systematic bias with respect to participant characteristics — not random error but structured distortion that required locally trained, human-validated alternatives to detect and correct. Morgan (2023), testing ChatGPT against two previously analysed qualitative datasets, found that while the tool reproduced descriptive themes with reasonable fidelity, it consistently missed subtle interpretive material and required embedding within a larger analytic process — but provided no infrastructure to support or document that embedding. None of these observations are new to researchers who have tried to use existing tools for sensitive or methodologically demanding work. What is absent is an architectural response to them — a gap that Toney and colleagues (2025) characterise directly as a failure of infrastructure rather than ethics: researchers are turning to AI because the tools that would truly support qualitative inquiry do not yet exist. The urgency of that absence is sharpened by recent demonstrations that agentic AI systems can, in principle, execute end-to-end research pipelines across many specialist tasks (Zhang, 2026). If the capability side of AI-assisted research is advancing rapidly, the governance infrastructure needs to keep pace — and Zhang's own analysis of where tacit human judgment remains irreplaceable makes explicit why that infrastructure cannot be informal.

## Existing Tools and Their Limitations

A survey of the current landscape is necessary background, because the pipeline described here makes most sense as a response to specific limitations in what already

exists rather than as an abstract design exercise. We focus on the governance architecture of existing tools rather than their analytical capabilities, because that is where the relevant comparison lies.

Commercial qualitative AI platforms such as CoLoop, Qualzai, and the AI-enhanced versions of established QDA software (NVivo 15, ATLAS.ti 25, MAXQDA AI Assist) share a broadly consistent model. The researcher uploads data to cloud infrastructure, specifies a task (code this transcript; identify themes across this corpus; suggest categories for these open-ended responses), and receives AI-generated output that can be reviewed and accepted or modified. These platforms vary in their sophistication, user interface quality, and the degree to which they allow the researcher to guide the analytical process, but they converge on the same fundamental architecture: cloud processing, no pre-registration linking, no documented researcher positionality, no programmatic audit trail, and no hard separation between the AI-generated output and the researcher's subsequent modifications to it.

The open-source landscape is thinner and less integrated. Tools such as iQual (Ashwin et al., 2022), which scales interpretive coding across large text corpora using supervised models trained on human-annotated examples, address specific analytical tasks competently but are not designed as research governance systems. There is no open-source tool that combines local model execution, reflexivity elicitation, two-pass analysis sequencing, and audit bundle generation in an integrated pipeline.

What unites both the commercial and open-source landscape is the implicit assumption that governance is the researcher's responsibility, and that the tool's job is to make the analysis faster and more capable. This is a reasonable division of responsibility for experienced researchers working with non-sensitive data in contexts where methodological conventions are well established and externally audited. It is significantly less reasonable when the researcher is a doctoral student learning qualitative methods for the first time, when the data requires GDPR Article 9 protections, or when the analysis is expected to meet the standards of a specific epistemological tradition that places explicit demands on reflexivity and evidence grounding. These are not edge cases; they describe a large proportion of qualitative research in health, education, psychology, and social work.

## Pipeline Design: Governance-First Architecture

### *Design principles*

The pipeline rests on four design principles that are worth stating explicitly before the architecture is described, because they explain why the implementation looks the way it does.

The first principle is that methodological safeguards should be enforced rather than recommended. A runbook or best-practice guide that tells researchers to lock their Pass 1 output before conducting the second pass, or to complete a DPIA before

ingesting sensitive data, relies on researcher compliance under conditions — time pressure, cognitive overload, unfamiliarity with the method — that are precisely the conditions under which compliance is most likely to fail. The pipeline treats these requirements as hard architectural constraints: the second pass cannot be initiated until a cryptographic anchor has been created for the first; the DPIA gate is a blocking check, not a reminder; the synthesis stage cannot proceed until every AI-generated claim has a completed human verdict record. This is not distrust of researchers — it is recognition that distributed, automated enforcement is more reliable than individual vigilance.

The second principle is that the researcher's positionality must be structurally embedded in the analysis, not appended to it afterwards. The structured reflexivity elicitation module — the lens dialogue — is not a questionnaire to be completed before analysis as a formality. It is a substantive, hashed component of the analysis record whose contents are used to configure the second analytical pass and whose relationship to the pre-registration record is explicitly checked. If a researcher introduces hypotheses in the lens dialogue that were not in their pre-registration, those hypotheses are flagged as post-hoc in the audit trail. This creates an incentive structure for honest pre-registration rather than relying on researcher integrity alone.

The third principle is that local execution is the default, not an option. For studies involving special-category data under GDPR Article 9, sending participant data to an external API is not a governance choice that can be left to researcher discretion — it is a legal processing question that must be resolved before analysis begins. The pipeline runs on locally hosted LLMs (via Ollama or llama.cpp) on hardware that the researcher controls. This makes local execution the path of least resistance rather than the technically demanding alternative.

The fourth principle is that the pipeline should be method-agnostic within a governance-consistent framework. The specific analytical modules — the IPA initial noticing agent, the discourse feature extraction agent, the reflexive TA coding agent — are configured by the researcher’s method declaration at onboarding. The governance layer (DPIA gating, two-pass sequencing, grounding verification, audit bundle generation) is identical regardless of method. Crucially, the method-routing layer does not merely accept a method declaration passively — it actively proposes the most appropriate analytical approach or combination of approaches for the research question and data type presented, explains the methodological rationale for its suggestion, and then configures the pipeline accordingly. Where the researcher wishes to specify a different method or adjust the routing, the pipeline optimises around those preferences; the researcher’s methodological authority is always final. This guidance function is particularly valuable for researchers with less formal training in qualitative methodology, for whom selecting between, say, reflexive thematic analysis and interpretive phenomenological analysis is not a straightforward decision. This means the same infrastructure can support a doctoral student conducting interpretive phenomenological analysis of clinical interviews and a public health team

running thematic analysis across a large survey corpus, without compromising the methodological commitments of either.

### *The two-pass analysis structure*

The pipeline's most distinctive structural feature is the enforced separation between a blind first pass and a theoretically positioned second pass. This design addresses a specific methodological problem: when a researcher asks an AI to analyse data with their theoretical frame already in place, there is no way to distinguish what the AI found from what it was, in effect, directed to find. The researcher cannot know whether the themes that emerged were genuinely present in the data or whether the AI amplified signals that matched the provided vocabulary while suppressing signals that did not. The two-pass structure makes this problem tractable.

The first analytical pass is conducted without researcher context. The AI reads the de-identified data and produces an initial analytical output — candidate themes, notable features, patterns across participants — using a system prompt that is deliberately free of theoretical framing. This output is immediately hashed (SHA256) and the hash is deposited to an external immutable record (OSF deposit or institutional repository) before the researcher sees it. The output cannot be modified after this point without the modification being detectable from the hash mismatch, and the external deposit creates a tamper-evident record independent of the researcher's local files. This is the Pass 1 anchor.

Only after the Pass 1 anchor is confirmed does the researcher engage in the lens dialogue — a structured ten-question interview that elicits their theoretical orientation, working hypotheses, expected findings, relevant vocabulary, and interpretive boundaries. The lens dialogue output is saved as a signed JSON record whose hash is linked to the researcher's identifier and timestamp, creating an auditable record of the researcher's stated positionality at the moment the second pass begins. Any hypothesis stated in the lens dialogue that was absent from the pre-registration record is automatically flagged as post-hoc in the audit trail.

The second analytical pass reads both the de-identified data and the lens summary. The system prompt injects the researcher's stated theoretical orientation and directs the AI to re-analyse the data through that lens, while also explicitly requiring the AI to surface material that resists or complicates the lens — what the pipeline calls the disconfirmation mandate. The second pass produces a delta report comparing Pass 1 and Pass 2 outputs, making the interpretive effect of the researcher's lens empirically visible rather than simply asserted.

### *Grounding verification and human evidence review*

A persistent concern about LLM-generated qualitative analysis is hallucination: the production of plausible-sounding claims that are not, in fact, grounded in the data provided. For a research context where analytical claims need to be linked to specific participant statements, this is not an acceptable failure mode. The pipeline addresses

this through a dedicated grounding verification pass conducted after the second analysis.

The grounding verification agent takes the Pass 2 output and the original de-identified transcripts and, for each claim, identifies and verifies the supporting evidence: does the cited excerpt exist at the stated location? Does it actually support the claim as stated? How many distinct participants and excerpts support this claim? Claims are rated on a four-level support strength scale (strong: three or more excerpts from three or more participants; moderate: two; weak: one; none: zero or unverifiable) and flagged with specific verification issues — overgeneralisation, excerpt-does-not-support, single-participant-only, and others. Crucially, the grounding agent does not modify claims; it only flags them. The records it produces have the human_verdict field set to null.

The pipeline supports, and recommends, running this verification pass using a model from a different provider and training lineage than the primary analysis model. The rationale is methodological: a model that generated a claim shares representational blind spots with a verifier from the same training run, making self-verification unreliable in exactly the cases where it matters most — where the primary model has systematically misread or over-interpreted a passage. A verifier from a different training corpus and architecture is more likely to surface these errors because its failure modes differ. This mirrors the principle of independent review in human research, and it mirrors the process by which this pipeline was itself developed and reviewed. The pipeline's configuration layer allows the verification model to be specified independently of the analysis model.

The human evidence review is the final non-delegable step before synthesis. A researcher reviews every flagged claim, sets a verdict (accept, accept with revision, reject, or recheck pending further examination), and provides notes for any revision or rejection. Claims that are accepted despite weak or absent evidential support are automatically labelled as interpretive propositions in the audit record, and the pipeline requires a specific template sentence to be included in the write-up for any such claim. The synthesis stage is blocked until every verdict field is complete. This is the mechanism by which the pipeline prevents AI-generated claims from entering the findings without explicit human accountability for each one.

The pipeline further distinguishes between two types of claim requiring heightened scrutiny. An interpretive proposition, as described above, is a claim retained despite weak or absent evidential support — its evidential fragility is the concern. A theory-sensitive claim is a different category: one that introduces a novel theoretical construct, a cross-case generalisation, or an interpretive frame that extends beyond direct data description, regardless of how many excerpts it cites. Theory-sensitive claims are detected heuristically by the grounding verification agent and flagged for a separate theory sign-off step, in which the reviewer must provide a human-authored theoretical justification (not AI-generated) before the claim may enter any public artifact. This distinction matters because the risk profile differs: the problem with a theory-sensitive

claim is not that the evidence is thin but that the interpretive move is large, and large interpretive moves require explicit human ownership regardless of evidential support. This reflects Zhang's (2026) argument that theoretical originality is the locus of irreducible human value in research — the pipeline makes that principle structurally enforced rather than aspirational.

A further component of the governance layer addresses disclosure. The pipeline generates a filled authorship disclosure template as part of the audit bundle, ready for inclusion in any resulting methods section. The template documents the modules used, the human verification steps performed, the pre-registration DOI, and the audit bundle deposit location — producing this document automatically from the run record removes a significant friction point, since researchers know they should disclose AI usage but often struggle with the appropriate level of granularity. A pre-filled template grounded in the actual analysis is more likely to result in complete and accurate disclosure than a general injunction to be transparent.

### *Stability testing*

Any LLM-generated analysis raises questions of reproducibility: would a different run of the same model, or a run on a different model, produce the same findings? The pipeline addresses this by running Pass 2 four times — three runs with different random seeds and one fully deterministic run at temperature zero — and computing a set of stability metrics across the outputs. The theme stability score measures the proportion of themes that are consistent across all four runs. The Jaccard overlap score measures set overlap of theme labels across run pairs, providing a quantitative index of how much the analysis drifts across seeds. The lens amplification index measures the proportion of new observations in Pass 2 (absent in Pass 1) that are supported only by vocabulary drawn from the researcher's lens — a signal that the AI may be reflecting the researcher's frame back rather than finding independent evidence for it. These metrics are included in the audit bundle and are available for inclusion in the methods section of any resulting publication.

## What the Architecture Makes Possible

The governance architecture described above is not primarily about adding complexity to qualitative research — it is about making a set of existing methodological commitments implementable by researchers who would otherwise struggle to meet them. It is worth being concrete about who those researchers are and what the pipeline changes for them.

A doctoral student conducting semi-structured interviews with mental health service users for an IPA study faces several simultaneous challenges that current tools cannot help with: the data is GDPR Article 9 special-category material that cannot be processed through cloud services; the method requires documented reflexivity; the institution will expect an audit trail for the viva; and the student is, by definition, learning the method rather than practising it expertly. The pipeline addresses each of these

without requiring the student to be an expert in GDPR law, cryptographic hashing, or the epistemological distinctions between reflexive and positivist qualitative approaches. The DPIA gate enforces the data governance before analysis can begin; the lens dialogue elicits and documents reflexivity in a structured way that meets Lincoln and Guba's (1985) confirmability criterion; the audit bundle is the audit trail. The student's time is freed for the interpretive work that is the actual methodological contribution.

A small research team in a clinical or applied psychology setting, working under time pressure on a mixed-methods study combining therapy session transcripts with questionnaire data, faces a different but related set of challenges: the team needs to be able to demonstrate inter-coder reliability (for the quantitative strand) and reflexivity (for the qualitative strand) while coordinating across multiple researchers who may have different theoretical orientations. The pipeline's team workflow activates an adjudication module when inter-coder Krippendorff's alpha falls below 0.667, requiring documented resolution of disagreements before synthesis proceeds. The method-routing layer applies the appropriate quality criteria to each strand automatically.

A researcher working in a community or voluntary sector context, without institutional infrastructure, access to QDA software licences, or a large support team, faces a capacity problem that has historically meant either limiting the scope of the analysis or accepting that the level of methodological rigour achievable is lower than it would be in a well-resourced university setting. The pipeline runs on hardware now available at consumer price points — a machine with a modern consumer GPU or mixed CPU–GPU configuration depending on VRAM, is sufficient to run the local models required — and the entire infrastructure is open-source and free to use. The governance architecture means that the analysis produced by a solo researcher with limited institutional support can be documented and defended to the same standard as one conducted by a larger team.

There is a further equity dimension that current tools do not address. Qualitative research methodology, as traditionally taught and practised, places significant demands on working memory, the ability to manage non-linear multi-stage processes simultaneously, and the capacity to maintain consistent documentation across extended periods of analysis. These demands can create substantial barriers for researchers with dyslexia, ADHD, autistic cognition, or other forms of neurodivergent processing — not because those researchers lack the interpretive or theoretical ability that qualitative analysis requires, but because the procedural complexity of the method as conventionally executed can be genuinely prohibitive. The pipeline structures that complexity into enforced sequential stages with explicit checkpoints, automated documentation, and clear decision gates, separating the procedural demands of rigorous analysis from the interpretive and theoretical work that is the actual intellectual contribution. For researchers who find the conventional multi-threaded qualitative workflow cognitively inaccessible, this structural scaffolding can make the

difference between being able to conduct the work at all and being excluded from a methodological tradition they are otherwise fully equipped to contribute to.

There is also a less obvious beneficiary: the qualitative research community's ongoing conversation about what rigour looks like in the age of AI. One of the most productive questions raised by AI-assisted analysis is precisely the question the two-pass structure makes empirically tractable: how much of what a researcher 'finds' in qualitative data is a function of the data, and how much is a function of the theoretical lens through which it is approached? The Pass 1 versus Pass 2 delta, combined with the lens amplification index, generates data about the relationship between positionality and analytical output that has previously been accessible only through painstaking manual reflexive analysis. Used at scale across multiple studies, this could contribute substantially to methodological conversations about the nature of interpretive knowledge.

The architecture also creates a principled pathway for responsible scaling. A rigorous deep analysis of a small sample — conducted with full human review of every claim — produces more than findings: it produces a calibrated interpretive framework, validated coding categories, and a stability baseline for the domain and model configuration. The pipeline is designed to support a sequential two-round research design in which this framework informs a larger follow-up study with a stratified review protocol: theory-sensitive and weakly evidenced claims receive full human review regardless of sample size, while strongly evidenced claims are sampled at a declared rate specified in the pre-registration. This is not automation of interpretation — the Pass 1 blind run, grounding verification, and OSF anchoring remain unchanged, and the epistemological status of any scaled analysis must be declared explicitly in the pre-registration. It is, rather, a redistribution of human effort toward the moments where tacit judgment is irreplaceable, and away from the moments where it is not. The infrastructure for this is present in the current release; the methodology for doing it responsibly is documented in the repository's scaling framework guide.

In keeping with the reflexive transparency advocated by the pipeline itself, it is worth noting that the design of this system emerged through a process of human–AI collaboration. The author is a postgraduate researcher in psychotherapy rather than a trained software engineer, who identified the governance gap described in this paper through extended engagement with agentic AI tools in research and exploratory contexts. The conceptual architecture — including the governance principles, methodological constraints, two-pass locking structure, and human-verification framework, multi-LLM passes for stability — was developed by the author. AI assistance was used primarily as an implementation aid, supporting the translation of these conceptual decisions into formal technical specification, repository scaffold, and test framework. This development process serves as a practical illustration of the paper's central argument: that effective human–AI collaboration in research depends not on replacing human judgment but on structuring workflows so that human

interpretation, theoretical framing, and critical scrutiny remain visible and accountable within the research record.

## Limitations and the Case for Empirical Evaluation

The most significant limitation of this paper is also its most important honest statement: nothing described above has been empirically tested. The pipeline is a design contribution, not an evidence-based recommendation. The claims made about what it makes possible are reasonable extrapolations from the design principles and the existing literature on AI-assisted analysis, reflexivity, and audit methodology — they are not claims demonstrated in live research contexts. Whether the two-pass structure actually improves the quality of qualitative analysis, whether the lens dialogue adequately elicits the positionality information needed to configure the second pass meaningfully, and whether the grounding verification agent achieves acceptable recall and precision in identifying unsupported claims are all open empirical questions.

There is also a question about the LLM at the centre of the pipeline. The pipeline is model-agnostic in the sense that any locally runnable instruction-following model can be used, but the quality of the analytical output clearly depends on the capability of that model. Recent locally runnable models have reached levels of instruction-following and reasoning capability that make structured qualitative analysis prompts viable on consumer hardware. The pipeline is model-agnostic and can be configured with any compatible local model.However, performance on qualitative analysis specifically — characterised by interpretive complexity, domain-specific vocabulary, and sensitivity to theoretical framing — has not been benchmarked in a way that is directly relevant to this pipeline’s use cases, and results will vary with model family, prompt engineering, and configuration. The relationship between local model capability, analysis quality, and the governance constraints enforced by the pipeline will need to be characterised empirically.

A further limitation is that the pipeline does not, and cannot, replace the interpretive expertise of the researcher. The system prompts for the IPA and discourse analysis modules are designed to activate the LLM's existing knowledge of those methods. Whether a prompted general-purpose LLM produces output that meets the epistemological standards of those traditions as understood by expert practitioners is, again, an empirical question. What the pipeline does is ensure that whatever the LLM produces is transparently documented, independently verifiable, and subject to human review before it is treated as a finding. That is a weaker claim than 'the AI does good IPA', but it is a claim that can be made honestly at this stage.

We invite the research community to evaluate the pipeline empirically, and have designed the repository with that invitation in mind. The documentation includes synthetic test fixtures that allow the pipeline to be run without live participant data, enabling evaluation of the governance mechanisms independently of the analytical quality. Researchers interested in piloting the system in live studies are encouraged to engage with the repository and to contribute findings back to the project. The design

is explicitly provisional — the architecture reflects our current best understanding of what responsible AI-assisted qualitative research requires, and we expect that understanding to be revised by evidence.

Reproducing this design. To run the pipeline against the included synthetic fixtures: (1) install Ollama and pull the target model (e.g. qwen3.5:27b as of March 2026); (2) clone the repository and install dependencies via pip install -r requirements.txt; (3) run make test-local to execute all governance tests against mock LLM outputs; (4) run python src/pipeline.py --demo to execute a full demo pass against the synthetic transcript fixtures. The demo produces a complete audit bundle including a Pass 1 hash, a pre-filled lens record, grounding verification output, and evidence review records. Seeds and model configuration are recorded in the audit bundle. The repository includes artifacts/audit_schema.json, examples/lens_example_locked.json, examples/dpia_signed.json, and a demo audit bundle, allowing independent inspection of all schema and governance constraints without running the pipeline.

## A Note on Publication Pathway for Untested Pipeline Work

We include a brief discussion of publication strategy because we believe the norms around publishing design contributions in research methods are less well established than they should be, and the ambiguity creates disincentives for exactly the kind of speculative infrastructure work that the field needs. A design paper that accurately describes an untested architecture and makes honest claims about what it might achieve is not a weaker contribution than an evaluation paper — it is a different kind of contribution, with different epistemic standards and different value.

The appropriate standard for a design contribution of this kind is: Is the gap it addresses real and documented? Are the design decisions well-motivated by reference to existing literature and identified problems? Is the artefact publicly available for inspection and evaluation? Does the paper make clear what is claimed and what is not? We believe this paper meets those standards. We intend to submit a preprint version to arXiv (cs.CY) to establish public priority and make the design available for immediate community engagement, and we are submitting this paper for peer review at the International Journal of Qualitative Methods, which has an established track record of publishing methodological design contributions alongside empirical findings.

## Conclusion

The relationship between artificial intelligence and qualitative research is going to develop whether methodologists engage with it or not. The tools are already in use — students are uploading interview transcripts to ChatGPT, teams are adopting commercial QDA platforms with AI layers, and the norms for what counts as acceptable AI assistance in qualitative work are being established in practice faster than they are being established in the literature. This is not a reason for methodologists

to be cautious about engaging; it is a reason to engage urgently, because the window for shaping the emerging norms is open now.

The pipeline described in this paper is one attempt to engage concretely with that challenge. Its argument is simple: the capability of LLMs to assist with qualitative analysis is real and growing, but deploying that capability responsibly requires a governance architecture that current tools do not provide. An architecture that enforces two-pass analysis sequencing, embeds researcher reflexivity as a signed and hashed record, gates special-category data processing behind a completed DPIA, and requires human sign-off on every AI-generated claim before it enters the findings does not guarantee good qualitative analysis — nothing can do that. But it does make the analysis auditable, transparent, and defensible in a way that the current generation of AI research tools does not.

We offer the pipeline as an open invitation: to inspect the design, critique the assumptions, and most importantly, to test whether it works.

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*Open Science Statement: The pipeline repository and full documentation are publicly available at https://github.com/ada3216/Human-Agentic-AI-Adaptive-Research-Pipeline-HAAARP. All design documents, architecture specifications, and development plan are included. No participant data is present in the repository. The code and this paper are released under the Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Commercial use is available under a separate commercial license by arrangement with the authors.*

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